

# Animesh Kumar Singh

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## WORK EXPERIENCE

- Facctum**
    - Associate ML Engineer*
      - Sanctions Press Release Extraction System:** Integrated large language models to extract critical regulatory entities from complex HTML/PDF press releases, mapping raw data to a **Common Data Format**. Delivered an **audit-ready**, delta-tracking compliance system that **improved** data extraction accuracy by **83%** and accelerated downstream regulatory **SLAs** targets to Banks by **94%**.
      - Trade Goods Screener:** Engineered an end-to-end intelligent Trade Goods Screener using **LangGraph** and **Bedrock's Claude Opus Model**, enabling HS-code classification, sanctions/license screening, risk scoring, and case routing to compliance officers. Built an orchestration layer with role-based memory and tool integrations, reducing manual review time by **70%** and removing coordination bottlenecks across trade compliance operations.
  - Volkswagen (VGDS - India)**
    - SDE Intern*
      - Agentic Service Center Automation :** Designed an end-to-end intelligent workflow for automotive service centers using **CrewAI** and **Bedrock's Llama-3B**, enabling autonomous task routing from UI intake to technician assignment, inventory checks, real-time communication, and payment processing. Implemented multi-agent orchestration with role-based memory using **tool-augmented agents**, reducing service resolution time by **45%** and eliminating manual coordination bottlenecks across operations.
      - Hybrid Chatbot System for In-Drive Personalization:** Engineered a hybrid recommendation model integrating **LLama 3.1 LLM** with a dynamic **RAG** retrieval mechanism for real-time, in-vehicle assistance. The system encodes longitudinal user preference data (k-past interactions) into semantic vectors for retrieval, providing rich, contextual grounding for the generative model. Model efficacy was validated by a significant reduction in perplexity and almost **28%** improvement in Mean Reciprocal Rank (**MRR**) for Point-of-Interest suggestions, proving high-fidelity predictive accuracy.
  - Medecro AI**
    - Machine Learning Engineer Intern*
      - Dental YOLOv8 Object Detector:** Developed a **YOLOv8**-based object detection model on Medecro.ai's annotated dental X-ray set (10k+ images) using Mosaic & CutMix enhancements. achieving **mAP@0.5 0.92**, and **25%** model size reduction via structured pruning & **8-bit quantization**, deployed as endpoint for real-time chairside diagnostics.

Pune

Sept 2025 - Present

Pune

Jan 2025 - July 2025

Nov 2024 - Jan 2025

## PROJECTS

- AI-Driven Cold Mail Generator:** Architected a generative outreach engine using Llama & **ChromaDB** with hybrid dense/sparse retrieval and FAISS indexing, **LoRA** fine-tuned LLMs, and optimized query pipelines for <100ms latency cutting email drafting time by **59%+** and driving a **30%+** lift in response rates.
- LLaMA8B Domain-Adaptation Pipeline:** Built a Colab-optimized fine-tuning setup for **LLaMA-3 8B** with 4-bit bitsandbytes quantization, **LoRA r=16 adapters**, and gradient checkpointing. Trained 10 epochs on 2048-token sequences (batch size 8), achieving 300tokens/sec throughput, **85%** GPU memory savings, and <50ms inference latency
- Credit Risk Assessment Engine:** Engineered an end-to-end credit risk prediction pipeline in Python, leveraging **Decision Tree, Random Forest, and XGBoost** classifiers on 10K+ customer profiles. Designed advanced feature engineering with financial ratio analysis, missing-value imputation, and hyperparameter optimization, achieving a **5%** lift in accuracy over industry baselines and enabling robust, real-time lending decisions.

## EDUCATION

- Bachelor of Technology - Computer Science and Engineering**
    - Silicon University (CGPA - 8.02)*
- 2021- 2025

Bhubaneswar, India

## TECHNICAL SKILLS

- Core Competencies:** Data-Science, Machine-Learning, Agentic AI, Deep Learning, Computer Vision, Python, MLOps
- Frameworks & Libraries:** TensorFlow, PyTorch, Scikit-learn, Pandas, NumPy, Matplotlib, Seaborn, Keras, XGBoost, Random forest, LightGBM, FastAPI, Flask, Streamlit
- AI & LLM Technologies:** OpenAI, Claude, Gemini, LLAMA2, BERT, Transformers, Langchain, HuggingFace, FLux, Mistral, Deepseek, Crewai, LangFlow
- Vector Databases & Search:** FAISS, Pinecone, Weaviate
- Data Engineering & Big Data:** PostgreSQL, MongoDB, MySQL, Oracle
- Cloud & DevOps:** AWS (EC2, S3, Bedrock, Sagemaker, Lambda)
- Software Engineering:** Git Version control, Docker

## ACHIEVEMENTS & CERTIFICATIONS

- Competition Performance:** Volkswagen imobilothon 4.0 - TOP 5
- Work Performance:** Star Performer - 2026 (Q2)